

Working with Images

Photo vs Web Images
Size Matters



3.2 MB (3,277 kb)
 $2,592 \times 3,872 = 10,036,224$ pixels

More than 10 million pixels to move!

47 kb
 $200 \times 300 = 60,000$ pixels

"Just because it is on the Web doesn't mean it is yours."

-- The Internet Police

Creative Commons

A work is either copyrighted or it isn't. Black and white. Yes or No. Binary.

You either owned it or you didn't.

That didn't work for some people.

So, they devised [Creative Commons](#), a licensing scheme that allowed the owner to share some of their rights with others.

Suddenly there are shades of gray allowing you to customize the license depending on how you want your work utilized by others.

You still have the copyright for your work, but now you can let other people:

- Use it for non-profit use or...
- Use it for profit as long as they don't change it or...
- Use it and modify it anyway they want for profit as long as you get credit for it.

For example, you take a really great photo and publish it out on [Flickr.com](#).

You set the Creative Commons license describing how the photo can be used.

License Features

Your choices on this panel will update the other panels on this page.

Allow modifications of your work?

☒ Yes ☐ No

☐ Yes, as long as others share alike

Allow commercial uses of your work?

☐ Yes ☒ No

License Jurisdiction:

International

Selected License

Attribution-NonCommercial 3.0 Unported

CC BY NC

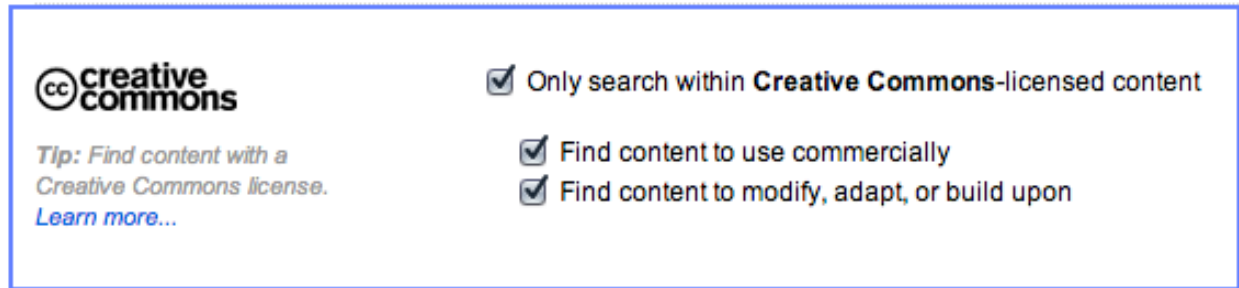
This is not a Free Culture License.

Need photos for your web pages?

Almost any image showing in a web browser can be saved to your computer. Simply right-mouse click on the image and select "save image as". But, just because you can save it to your computer doesn't mean it is yours to use. You must have permission from the owner.

Free Photos

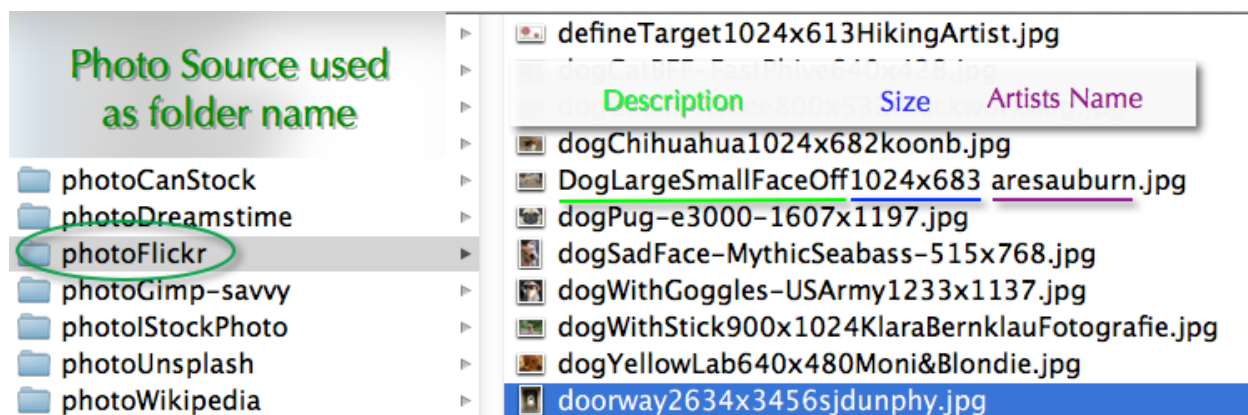
Free photos are available out on [Flickr.com](https://www.flickr.com) thanks to Creative Commons. Click on the search area and then select "Advanced Search". Near the bottom of the list of options you will find selections for Creative Commons. Check all the boxes in order to get the license with the broadest permissions.



Every creative commons license, no matter how broad, asks that you give the artist credit or attribution.

Save these photos in a special folder away from your web page development. A smart way to keep attribution credits is to save the photo using a descriptive phrase, the size of the image, and the name of the artist as part of the filename.

Here is an example showing photo folders for various sites and the naming convention used for each photo. (Notice how all the folders start with "photo" so they stay grouped together in the directory listing.)



These are your original photos. You will be making copies of these images, editing them for the Web, and saving them in the graphic folders of your web sites.

Stock Photos (\$\$\$)

There are several stock photo sites. On these sites you pay to use a photo for a specific license and normally for a particular application. For example you might have to pay a fee to use the photo on the Web and another fee to use the photo in a brochure. One of the most popular is iStockPhoto.com. Another site that has fairly lenient licensing (and prices as low as \$1.00 for a small photo) include DreamsTime.com and Unsplash.com

Make certain to read through the entire license for the photo you are using and use the credit line as specified.

Public Domain Photos

There are a lot of archive sites with older photographs that are now public domain. There is no licensing and you can use the photograph as you like. However, most of the photos are very old and don't normally work well for modern sites. The [US Library of Congress](http://www.loc.gov) has an extensive collection including photos from the Civil War up to the present. <http://www.flickr.com/commons> is a collection of collections and includes the Library of Congress photos as well as those from [the Smithsonian](http://www.smithsonian.gov).

Here is a photo from the Library of Congress taken by Dorothea Lange, a famous photographer during the Depression.



HTML is built using text files. Graphics are separate files. They are transported across the web along with the HTML text file and it is the browser that puts them both together.

The element

In order to tell the browser what picture to display use the element. Most web pages have lots of graphic images so it is smart to organize them in a sub-folder.

Set up a basic web page using your HTML Template

Add the element including the src=" " attribute to link the HTML code with the actual image file as well as the alt=" " attribute to give alternative text in case the image cannot be found or the page is being read by a audio browser or a search engine such as Google.

Notice that is a single element so you have to include a space and the stop slash "/" before the closing bracket ">".

Also, by including two line breaks
 you can move the text in the paragraph below the image.

```

1  <!DOCTYPE html>
2  <html>
3  <!-- imageDemo.html - Using graphics on a web page.
4      Author:
5      Written:
6      Revised:
7  -->
8  <head>
9      <meta charset="utf-8" />
10     <title>Image Demo</title>
11 </head>
12
13 <body>
14     <h1>Using Graphics on a Web Page</h1>
15
16     <p>
17         
19         <br /><br />
20         This is a paragraph.
21     </p>
22
23 </body>
24 </html>

```

Here is the code:

```

<!DOCTYPE html>
<html>
<!-- imageDemo.html - Using graphics on a web page.
    Author:
    Written:
    Revised:
-->
<head>
    <meta charset="utf-8" />

```

```
<title>Image Demo</title>
</head>

<body>
  <h1>Using Graphics on a Web Page</h1>

  <p>
    
    <br /><br />
    This is a paragraph.
  </p>

</body>
</html>
```

Here is what your output will look like:

Using Graphics on a Web Page



This is a paragraph.

Increase the SEO (Search Engine Optimization) of your page.

The alt=" " attribute is very useful in telling search engines such as Google the important keywords on your site. alt=" " attributes, headings, and the <title> text are all key indicators that Google and other search engines use to

determine what the web page is about (your keywords).

Instead of saying "Suzy Brown" add more information to the alt=" " text. For example, "Portrait of Suzy Brown, President of Web Art and Design"

Often, as part of the page design, you will want to wrap the text of the paragraph around the image. This can be done using CSS (Cascading Style Sheet).

In order to see the effect of wrap-around text go out to the Lorem Ipsum website: lipsum.com and add several paragraphs to the body of the page right after the element. You will probably want to remove the two

 you originally added.

Your code should look something like this:

```
<!DOCTYPE html>
<html>
<!-- imageDemo.html - Using graphics on a web page.
      Author:
      Written:
      Revised:
-->
<head>
  <meta charset="utf-8" />
  <title>Image Demo</title>
</head>

<body>
  <h1>Using Graphics on a Web Page</h1>

  <p>
    
    This is a paragraph. Lorem ipsum dolor sit amet,
    consectetur adipiscing elit. Sed vulputate justo sit
    amet arcu volutpat sed laoreet nulla luctus. Sed
```

viverra diam vitae quam sollicitudin rhoncus vitae at
eros. Donec gravida quam sem. Maecenas id magna nec
libero faucibus consectetur.

</p>

<p>

Phasellus consequat faucibus ornare. Integer arcu mi,
scelerisque sit amet dignissim non, luctus posuere
magna. Nulla odio est, pretium eget sollicitudin in,
dictum nec velit. Curabitur dapibus ullamcorper nisl,
in aliquam nibh rhoncus quis. Mauris commodo, elit
vitae convallis sollicitudin, leo massa ullamcorper
neque, a convallis ligula nulla ac tellus. Suspendisse
accumsan, purus sed blandit sollicitudin, elit odio
faucibus enim, lacinia tempus dolor odio vitae dolor.

</p>

</body>

</html>

In the <head> section add a new <style type="text/css"> element. And inside this element add a selector for the img elements on the page.

```

1 <!DOCTYPE html>
2 <html>
3 <!-- imageDemo.html - Using graphics on a web page.
4     Author:
5     Written:
6     Revised:
7 -->
8 <head>
9     <meta charset="utf-8" />
10    <title>Image Demo</title>
11    <style type="text/css">
12
13        img {
14            float: left;
15        }
16
17    </style>
18 </head>
19
20 <body>
21     <h1>Using Graphics on a Web Page</h1>
22
23     <p>
24         
26         This is a paragraph. Lorem ipsum dolor sit amet, con
          amet arcu volutpat sed laoreet nulla luctus. Sed vi

```

This CSS will style all the elements on the page.

Here is what the page will look like:

Using Graphics on a Web Page



This is a paragraph. Lorem ipsum dolor sit amet, consectetur adipiscing elit. Sed vulputate justo sit amet arcu volutpat sed laoreet nulla luctus. Sed viverra diam vitae quam sollicitudin rhoncus vitae at eros. Donec gravida quam sem. Maecenas id magna nec libero faucibus consectetur.

Phasellus consequat faucibus ornare. Integer arcu mi, scelerisque sit amet dignissim non, luctus posuere magna. Nulla odio est, pretium eget sollicitudin in, dictum nec velit. Curabitur dapibus ullamcorper nisl, in aliquam nibh rhoncus quis. Mauris commodo, elit vitae convallis sollicitudin, leo massa ullamcorper neque, a convallis ligula nulla ac tellus. Suspendisse accumsan, purus sed blandit sollicitudin, elit odio faucibus enim, lacinia tempus dolor odio vitae dolor.

I

Using Graphics on a Web Page



This is a paragraph. Lorem ipsum dolor sit amet, consectetur adipiscing elit. Sed vulputate justo sit amet arcu volutpat sed laoreet nulla luctus. Sed viverra diam vitae quam sollicitudin rhoncus vitae at eros. Donec gravida quam sem. Maecenas id magna nec libero faucibus consectetur.

Phasellus consequat faucibus ornare. Integer arcu mi, scelerisque sit amet dignissim non, luctus posuere magna. Nulla odio est, pretium eget sollicitudin in, dictum nec velit. Curabitur dapibus ullamcorper nisl, in aliquam nibh rhoncus quis. Mauris commodo, elit vitae convallis sollicitudin, leo massa ullamcorper neque, a convallis ligula nulla ac tellus. Suspendisse accumsan, purus sed blandit sollicitudin, elit odio faucibus enim, lacinia tempus dolor odio vitae dolor.

I

Notice how the text touches the image and how huge the image is?

You can adjust that using CSS.

```

10  <title>Image Demo</title>
11  <style type="text/css">
12
13  img {
14  float:    left;
15  margin:    20px;
16  width:     15%;      /* % of the browser window */
17  }
18
19  </style>
20  </head>
21

```

Here is the code for the <style> element. (Make certain this is inside of the <head> element on the page.)

```

<style type="text/css">
img {
float:    left;
margin:    20px;
width:     15%;      /* % of the browser window */
}
</style>

```

Here is what the page will look like:

Using Graphics on a Web Page



This is a paragraph. Lorem ipsum dolor sit amet, consectetur adipiscing elit. Sed vulputate justo sit amet arcu volutpat sed laoreet nulla luctus. Sed viverra diam vitae quam sollicitudin rhoncus vitae at eros. Donec gravida quam sem. Maecenas id magna nec libero faucibus consectetur.

Phasellus consequat faucibus ornare. Integer arcu mi, scelerisque sit amet dignissim non, luctus posuere magna. Nulla odio est, pretium eget sollicitudin in, dictum nec velit. Curabitur dapibus ullamcorper nisl, in aliquam nibh rhoncus quis. Mauris commodo, elit vitae convallis sollicitudin, leo massa ullamcorper neque, a convallis ligula nulla ac tellus. Suspendisse accumsan, purus sed blandit sollicitudin, elit odio faucibus enim, lacinia tempus dolor odio vitae dolor.

You can also float the photo to the right side of the page using float: right;

```
<style type="text/css">
img {
float:    right;
margin:   20px;
width:    15%;    /* % of the browser window */
}
</style>
```

This is the effect you will get. Resize the browser window to see how the text wraps around the image and the image size changes depending on the size of the browser window.

Using Graphics on a Web Page

This is a paragraph. Lorem ipsum dolor sit amet, consectetur adipiscing elit. Sed vulputate justo sit amet arcu volutpat sed laoreet nulla luctus. Sed viverra diam vitae quam sollicitudin rhoncus vitae at eros. Donec gravida quam sem. Maecenas id magna nec libero faucibus consectetur.



Phasellus consequat faucibus ornare. Integer arcu mi, scelerisque sit amet dignissim non, luctus posuere magna. Nulla odio est, pretium eget sollicitudin in, dictum nec velit. Curabitur dapibus ullamcorper nisl, in aliquam nibh rhoncus quis. Mauris commodo, elit vitae convallis sollicitudin, leo massa ullamcorper neque, a convallis ligula nulla ac tellus. Suspendisse accumsan, purus sed blandit sollicitudin, elit odio faucibus enim, lacinia tempus dolor odio vitae dolor.

Finally, if you want to have different borders for each side you can use all four parts of the margin: property

```
<style type="text/css">
img {
float:      left;
margin:     0px 20px 0px 0px; /* top right bottom left */
width:      15%;              /* % of the browser window */
}
</style>
```

The four numbers always represent in order: top right bottom left. (Think of a clock starting at midnight.)

This will remove the white space at the top and bottom of the image but put space on the right side between the image and the text.

Like this:

Using Graphics on a Web Page



This is a paragraph. Lorem ipsum dolor sit amet, consectetur adipiscing elit. Sed vulputate justo sit amet arcu volutpat sed laoreet nulla luctus. Sed viverra diam vitae quam sollicitudin rhoncus vitae at eros. Donec gravida quam sem. Maecenas id magna nec libero faucibus consectetur.

Phasellus consequat faucibus ornare. Integer arcu mi, scelerisque sit amet dignissim non, luctus posuere magna. Nulla odio est, pretium eget sollicitudin in, dictum nec velit. Curabitur dapibus ullamcorper nisl, in aliquam nibh rhoncus quis. Mauris commodo, elit vitae convallis sollicitudin, leo massa ullamcorper neque, a convallis ligula nulla ac tellus. Suspendisse accumsan, purus sed blandit sollicitudin, elit odio faucibus enim, lacinia tempus dolor odio vitae dolor.

JPG, GIF, PNG, and SVG Files

Most images used today contain millions of pixels. These graphic images have lots of repeating information. For example, a photo of a landscape might have row upon row of blue pixels in the sky.

To a computer scientist this is a huge waste of memory and data handling. Why handle each pixel separately when so many of them are the same. So, they came up with different ways to compress images into smaller files. This made it faster for computers to process and display each image.

Three of the most popular image compression formulas include JPG (pronounced: Jay-Peg), GIF (pronounced: jiff as in peanut butter), and PNG (pronounced: ping).

JPG files (.jpg or .jpeg) were designed to handle photographs. Images with lots of different shades and millions of colors.

GIF files (.gif) were designed for graphic logos and drawings with 256 colors or less. Later versions of GIF files came out that could handle more colors as well as transparency. Several GIF images could also be stored in the same file in order to create animated images. (Do a Google search for: animated gif generator. Here's one: <http://picasion.com/>).

PNG files (.png) are the newest type and can handle millions of colors as well as transparency. When in doubt use .png.

JPG, GIF, and PNG are all raster images or files based on individual dots or pixels.

Adobe Photoshop and GIMP are two popular photo editing programs designed for all three of these file types.

Just changing the file extension won't change the image type!

The file extension tells software how the pixels are arranged inside the file. The pixels in a JPG type file are arranged very different then a PNG file. Just changing the file extension won't change the file. It will only confuse the software trying to use that image.

Use a photo editor such as <http://pixlr.com/> , Gimp, or Adobe Photoshop to convert from one file type to another. The photo editing software knows how to rearrange the pixels for each file type you choose.

Why would you do this? Often times you may have a JPG file, say a company logo, that doesn't support transparency. By converting it to a PNG file type you can make the background transparent so it can be used with all types of backgrounds.

What is a BMP file?

BMP stands for BitMap and was used by Microsoft before the Web was invented. The JPG, GIF, and PNG files all compress/uncompress the pixels of an image making them smaller and more efficient to move across a network such as the Internet. A BMP file is not compressed. Every pixel stays in the file making BMP files big and bulky and not a good choice for web work. Whenever you come across a .bmp file use a photo editor and change it into a more efficient file type such as a PNG.

Another type of image is a vector. SVG files are vector files.

Unlike the pixel or raster images, a vector image is simply a text file of drawing commands that the browser uses to create the image. SVG (Scalar Vector Graphics) are excellent for drawings such as parts diagrams.

SVG images can be zoomed in without pixelating because each part of the drawing is rebuilt unlike the pixel structure of the other image formats.

The images on this site are all based on text SVG files <http://svg-wow.org/>

Adobe Illustrator and Inkscape are two popular programs used to create and edit SVG images.

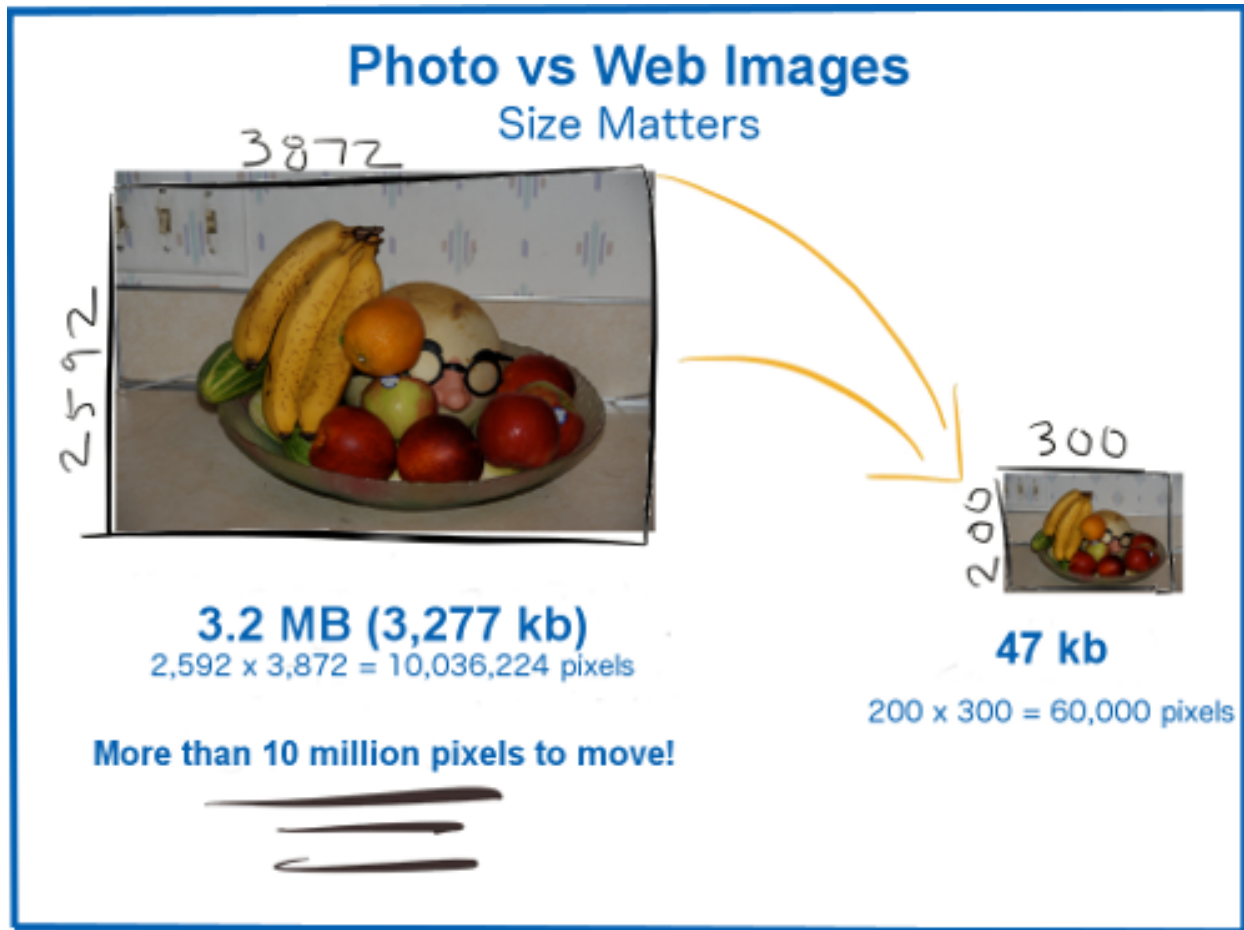
"You can make images smaller but it is illegal (and you will be punished) if you take small images and try to make them look bigger."

-- The Internet Police

Digital Cameras (including your mobile phone) create image files with lots of information. This allows images to be printed on photo paper in high resolution of 300 dpi (dots per inch) and larger than 11x14 inches. A printed high-resolution photograph needs LOTS of pixels.

A computer monitor on the other hand only will display 72 - 90 dpi. And, most often the image sizes are small like a wallet size photo. (Even smaller on a mobile phone.) So, a computer monitor doesn't really need all those extra pixels. And, people looking at web sites want them to load in three seconds or less.*

Don't slow down your web pages by using gigantic graphics. All those wonderful pixels will just go to waste while the user clicks off to another site.

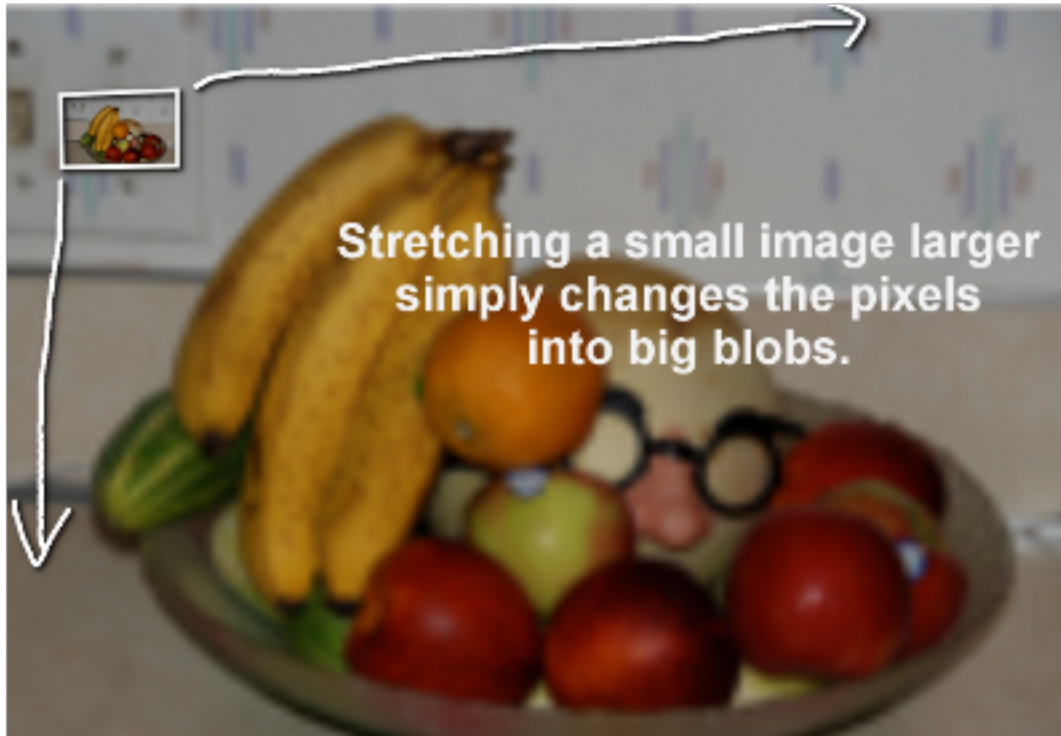


Best Practices

Save original images in a separate folder away from your web work.

You want to preserve the original images. Once you make an image smaller you've removed lots of information (lots of pixels) and you can't simply make the image bigger because the information is no longer there.

By keeping your originals separate you can always make a new, smaller copy for use on the Web.



Resize images to 300 pixels wide or smaller.

Change the resolution to 72dpi.

For your web photos use a photo editor to reduce the file size and resolution making the image as small as possible. Keep these "web ready" photos with your web work.

If a page is 800 pixels wide then a photo that is 300 pixels will fill almost half of the page.

Most images on a web page are 100 - 300 pixels wide.

There is a free online photo editor at <http://pixlr.com>. Use the express version to quickly resize an image. (The advanced version is very much like Adobe Photoshop). Or, do a Google search for: online photo editor

You can also use Gimp (free and open-source) and Adobe Photoshop (used by professionals around the world.)

Follow the naming conventions for all your image files including file extensions.

One of the most common problems with web work is images that don't display out on the Web. Web servers are case sensitive so it is important that you name your image files consistently.

Here are the naming conventions that will give you consistent filenames all the time. Use these for your files and folders.

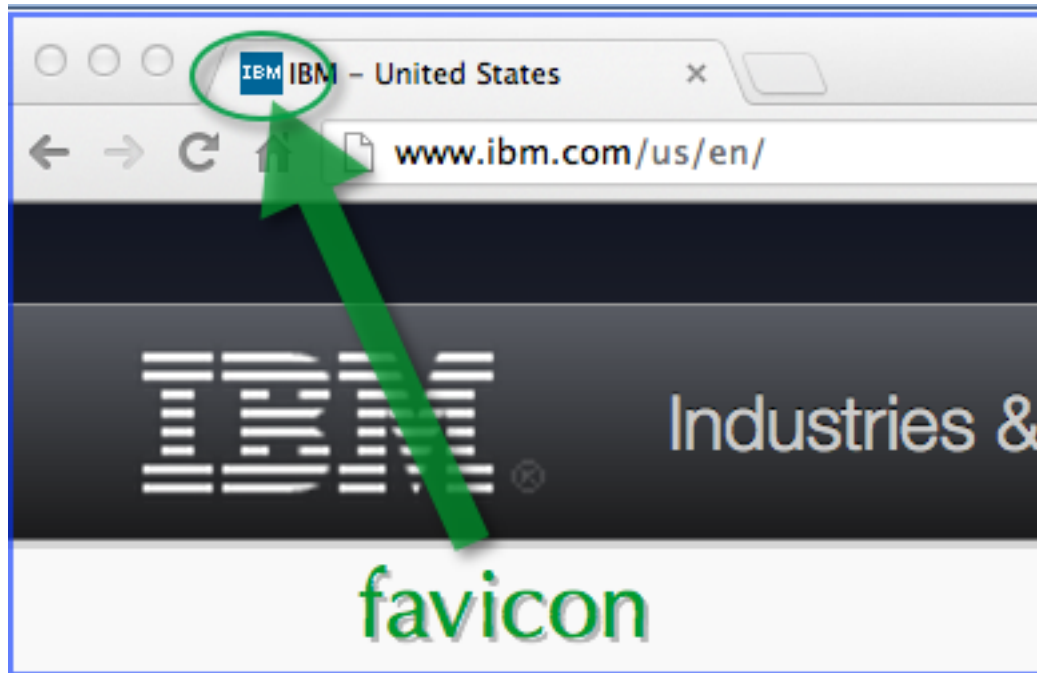
1. Always start with a lower case letter - UPPER/lower case matter. This keeps you from guessing, "Was it PhotoOfJoe.jpg or photoOfJoe.jpg?"
2. Don't use spaces. The Web hates spaces and changes them to %20. Pretty ugly!
3. Don't use underscores "_". If the filename is used in a URL (web address) the underscore will look like a space because most links are underlined.
4. useCamelCaseForReadability. By capitalizing the first character of every word your names will be very readable.
5. Keep everything singular. Don't make yourself wonder, "Was it graphics or graphic???" and then have to waste the time to look up the actual name.
6. All file extensions must be lower case. Often a graphic program will save a file using all caps. For example: myPhoto.JPG If you use this HTML `` the image will not display on the browser. UPPER/lower case matter.

*According to [surveys done by Akamai and Gomez.com](#), nearly half of web users expect a site to load in 2 seconds or less, and they tend to abandon a site that isn't loaded within 3 seconds. 79% of web shoppers who have

trouble with web site performance say they won't return to the site to buy again and around 44% of them would tell a friend if they had a poor experience shopping online.

Have you ever noticed those tiny icons next to the web address field in your browser?

Here's one that IBM uses on their site:



A favicon (.ico) is a icon-type image file that can be generated using online tools from other graphic images.

Here is a page where you can draw or upload an image to create a favicon for your own web pages:

<http://www.favicon.cc/>

You can name your file anything you like although by convention most people use favicon.ico.

Save your favicon.ico in the same folder as your web page.

On your web site add this code:

```
<link rel="shortcut icon" href="favicon.ico"
```

```
type="image/x-icon" />
```

Note: If you want to store the .ico file with the other graphic files you can put it in your graphic folder. Just make sure to include that folder name as part of the path:

```
href="graphic/favicon.ico"
```

When the browser finds this link element it will automatically add the small icon next to the URL of your page.

Tip:

The best favicon is a simple favicon. There's only 256 pixels in a favicon (16 wide x 16 high) so don't expect to be able to have your wedding photo convert ;-)

Summary

You now have the basic skills necessary for using all types of images on your web pages.

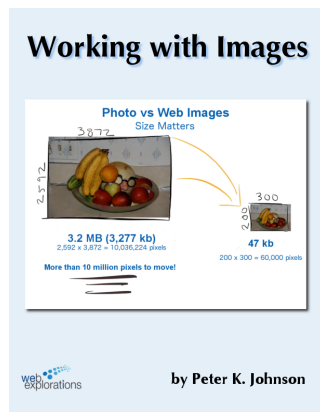
Some of the things you've learned include:

1. What Creative Commons is and how it allows you to use other people's photos with their legal permission.
2. Different places where you can find images for your web pages.
3. How to add images using the `<img />` element.
4. Why the `alt=""` attribute is so important for the SEO of each web page.
5. How to wrap text around an image.
6. How to place an image on the right or left side of a page.
7. Using CSS to style the `<img />` elements on a web page.
8. The different types of images: JPG, GIF, PNG, SVG, and (uncool) BMP.
9. Why small file size is important and how to make your images smaller for your web work.
10. How to add favicons to the URL of your web pages.

Credits

Working with Images, A tutorial written by [Peter K. Johnson - Web Explorations, LLC](http://WebExplorations.com) is licensed under a [Creative Commons Attribution-NonCommercial 3.0 Unported License](https://creativecommons.org/licenses/by-nc/3.0/).

Peter K. Johnson - <http://WebExplorations.com>
July 2013



© 2013, Peter K. Johnson